



PRODUCTION FACILITY & OUTPUT AUDIT REPORT

For PURO.EARTH

Company / CO ₂ Removal Supplier	Clean Maine Carbon, LLC
Production Facility Name	Greenville Facility
Production Facility Address	185 Greenville Steam Rd, Greenville, Maine, 04441
Production Facility Coordinates	45°45'57"N -69°58'77"W
Net Volume of CO ₂ Removal	114.3 CORCs
Removal Method	Biochar
Removal Period	August 2023 – August 2024
Auditors	Bill Chatterton Tim Hansen 350Solutions, Inc.

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2024



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PRODUCTION FACILITY & OUTPUT AUDIT REPORT

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Removal Method: Biochar		
Report Date: 10-30-2024		
Document No: 350VP-CMC-PU2404		
Rev: 2.1		

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1. INTRODUCTION

350Solutions was contracted to perform an audit and validation of the production facility as well as verification of carbon dioxide removal credit (CORC) claims for Clean Maine Carbon's (CMC) biochar production and utilization. CMC utilizes a slow pyrolysis process at its Greenville, ME facility to produce biochar from woody biomass from regional forest waste.

Regional forestry management and lumber production operations provide sustainable process feedstock streams to support CMC operations. CMC's use of these residual by-product streams to produce biochar presents a higher value use that leads to carbon sequestration. CMC currently supplies biochar as a soil amendment application to several end users.

350Solutions conducted an audit of the process, lifecycle CO₂ emissions assessment (LCA), and other administrative details to verify compliance with the requirements of the Puro.Earth Standard General Rules V3.1 [1] and Biochar Methodology (Edition 2022 Version 3) [2]. The audit and verification included a document review, a detailed data audit, and a site visit on September 27th, 2024.

TABLE 1. CMC VERIFICATION SUMMARY

Verification Summary	
CO₂ Removal Supplier	Clean Maine Carbon
Production Facility Name	Greenville Steam Facility GSRN: 643002406801000909
Removal Method	Biochar – Production from woody forest waste and use as soil amendment
Claimed CORCs	114.3 CORCs
Verification Report Date	10-30-2024
Site Visit Date	09-27-2024
Verification Period	08-14-2023 to 08-18-2024
Production Facility Location (Address and GPS Coordinates)	185 Greenville Steam Rd, Greenville, Maine 34°22'45"N 081°46'07"W
Verification Type	Combined Production Facility Audit and Output Audit for Puro.Earth, including on-site visit and facility audit; Puro Standard General Rules (v.3.1) and Annex A: Biochar Methodology (v3)

2. TECHNOLOGY DESCRIPTION

CMC operates a biochar production technology developed by BioMass Energy Techniques, Inc (BET), an original equipment manufacturer that owns the intellectual property for the BET BioMass Energy Systems. CMC utilizes a model BET 24PRD (pyrolysis rotary kiln). The rotary kiln is a pyro-processing device used to raise materials to a high temperature in a continuous process. Rotary kiln systems work by processing material in a rotating drum at high temperatures for a specified retention time to cause a physical change or chemical reaction in the material being processed. An aerial view of the facility in Greenville ME is shown in Figure 1.



FIGURE 1. AERIAL VIEW OF THE CMC GREENVILLE FACILITY

As shown in Figure 2 below, the biochar manufacturing process is comprised of a combustion chamber (A), a retention chamber (B), a rotary kiln (C), a thermal oxidation chamber (E), and a water cooler. The combustion chamber A is where wood fuel is fed for heating of the process and primary combustion takes place. Up to 2,200 pounds per hour (lb/hr) of biomass (at an assumed moisture content of 40%) is gravity fed into this chamber. Chamber B is a high-temperature retention chamber which allows condensable gases to achieve complete combustion. A negative draft pulls the gases off the feedstock in Chamber A and into Chamber B.

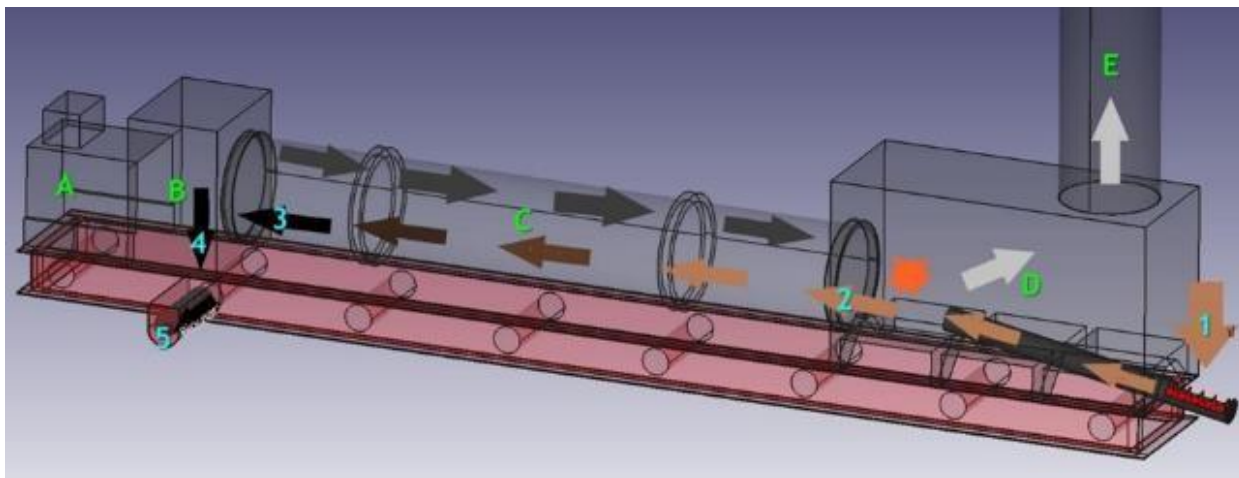


FIGURE 2. CMC'S BIOCHAR PRODUCTION PROCESS SCHEMATIC

Chamber C is a refractory-lined rotary drum pyrolysis kiln, in which thermal energy from Chamber B makes contact via countercurrent flow with the biomass feedstock (input into the process at point 1 in the diagram). Gases generated during the pyrolysis process in Chamber C increase the thermal energy of the system above the thermal energy in Chamber B. Chamber D is a thermal oxidation chamber that provides space and residence time to facilitate combustion of gases generated in Chamber C. Chamber E is the exhaust stack. CMC anticipates possible future scenarios that involve using the exhaust gases for other purposes including facility heating or power generation.

During typical operations, up to 1,000 lb/hr of green biomass feedstock is fed into the rotary drum at Point 1. An auger conveyance system moves the feedstock materials into the Chamber C rotary drum. The rotating drum moves the material towards the high temperature retention Chamber B exposing it to extreme heat which serves to complete the pyrolysis process. The size and rotational speed of the Chamber C rotary drum determines the retention time and can be adjusted by the operator. The feedstock material reaches its desired state, following pyrolysis at Stage 3, and drops down into the discharge area in Stage 4. This process is fed into a water cooler and exits the system at Point 5. There is also an additional discharge point biochar for the biomass used to heat the drum, which is seen in Figure 3.

Currently, CMC acquires feedstock from one regional lumber company. The feedstock material is waste hardwood from lumber operations (primarily oak, ash, and maple), processed into chips of 2 inches or less at the lumber mill. This feedstock is transported to the CMC facility in Greenville by truck with typical payloads ranging from 10-30 tons per load from sawmill residuals. The lumber mill currently has no other use for this waste material. The actual quantities, loads and sources for all feedstock delivered to the facility are tracked via trip tickets.

Feedstock stockpiled at the site is managed and fed into the biochar process by a diesel fuel powered front-end loader. The actual quantity of fuel used by this equipment is metered at an on-site fuel tank and equipment filling station and recorded on a weekly basis. The biochar production process is typically operated 24 hours per day, 5 days per week.



FIGURE 3. CMC'S BIOCHAR PRODUCTION PROCESS

During normal operations, any supplemental energy demands of the process required to meet proper operating temperatures are provided by biomass fuel that is a subset of the feedstock used for biochar production. For plant startup, the combustion chamber is charged with wood fuel and ignited using a propane torch. The quantity of propane used for each startup is recorded at the on-site propane storage tanks. The plant's process equipment utilizes electricity supplied by the regional electric grid. Electricity consumption of the biochar production process is not currently metered independently. Instead, conservative estimates of power consumption are reported based on readings from the revenue-grade, utility-owned meter installed at the facility. Meter readings are recorded at the utility meter with and without biochar production operations. These data allow operators to calculate a conservatively based estimate of the power consumed by the biochar production equipment and supporting electrical loads (lighting and controls).

Biochar leaving the pyrolysis kiln is quenched using potable water supplied by the local utility. The quantity of water used is estimated on a weekly basis. Critical process parameters to assure consistent, high-quality biochar include biomass feed rate, process temperatures, and pyrolysis residence time. Each of the key parameters are continuously monitored to assure consistent operating conditions.

After quenching, biochar is conveyed to a packaging process where it is placed into super sacks for storage and shipping. Each super sack contains nominal one ton of biochar at approximately 15 to 30% moisture content to ensure safe storage. Sacks of biochar are loaded onto trucks for delivery to regional farms for use as an agricultural amendment. End users typically apply up to 3 tons of biochar per acre using the same tractor-towed equipment that they use for application of other fertilizers and land treatments.

CMC collects biochar samples on a weekly basis for proximate analyses provided by a qualified laboratory. To date, two biochar samples have been analyzed according to the thresholds defined in voluntary quality standards for biochar, namely the International Biochar Initiative (IBI) Certification Program, Test Category A Parameters. CMC also performs on-site biochar moisture determinations every 4 to 5 hours during production.

3. AUDIT SUMMARY

3.1 AUDIT APPROACH

A planned series of audit activities were conducted by 350Solutions to independently validate and verify the production facility, its operations, production, and output data, and CORC claims. The audit was conducted following the specifications of Puro General Rules (v3.1) and the Biochar Methodology (v3). Specific audit activities conducted are summarized in Table 2. A completed Puro Biochar Methodology Compliance Checklist used during the verification is attached to this report as Appendix 1: Puro.Earth Biochar Methodology Audit Checklist. Verifier qualifications are attached as Appendix 3.

TABLE 2. AUDIT ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed*
09/12-26/2024	Introductory Document Review	- Review of LCA - Review of Puro CORC calculations - Review of facility registries and permits	- Puro_Registration_643002406801000909.pdf - Articles of Conversion to Clean Maine Carbon RS.pdf - Audit Document Index - Biochar - Review 4.xlsx - CMC LCA Projections Report_Final.docx - CMC GHG LCA Workbook updated_Final.xlsx - CORC Report Summary - Biochar_Final_v2.xlsx - A1164AN.pdf (air permit) - Report - CMC Air Emissions Test Report -Oct 2023.pdf - Environmental Evaluation Report_CMC_updated July 2024.pdf - GreenvillePermitLetter.pdf - 01_Stakeholder Engagement Report_CMC_04182023.pdf
09/12-26/2024	Data Review	- Review of waste biomass sources and sustainability - Review of system inputs and outputs - Review evidence of product output - Review of biochar properties - Review of biochar end use	- Additionality Form Signed.pdf - Clean Main Carbon - Cost Analysis - Financial Additionality (5.3.2023) (3).xlsx - Puro additionality questions to suppliers v1.8_CMC.docx - Biomass types and origins list _ CMC.xlsx - Letter from Feedstock Supplier.pdf - Summary of Forest Management in Maine.pdf - Feedstock invoices.pdf - Mass and energy balance of production process 5-14-24.xlsx - Biochar production equipment questionnaire 5-6-24.xlsx - Various facility digital graphics and video files - CMC_BiocharTestingProcedure_Final.docx - CMC_BiocharMoistureProcedure_Final.docx - End user invoices.pdf - 3060579-01 IBI Biochar Report.pdf - 3070467-01 Proximate Biochar Report.pdf - Deriving Soil Temperature for Biochar Application Using Invoice Addresses.docx - Soil Temperature Selection.png - Biochar Safety.pdf - CMC Carbon Pilot platform account; https://carbonpilot.vercel.app/company/output-report/overview
09/27/2024	On-site Visit	- Opening meeting and process walk through - Witness of operations, measurement points, and instrumentation - Review of equipment and calibrations	- Verifier observations of operations, measurement points, and instrumentation (see compliance checklist) - Observation and documentation of feedstock receipt, production logs, equipment utilized, data collected, material handling equipment and procedures - Product Weight Verification_CMC.pdf - Electricity Usage Records_scan.png

		- Review of intake and production data collection - Confirmation of company and facility administrative details - Confirmation of facility environmental and social safeguards	
10/04/2024	Post-visit Request for Information	Follow-up request for information sent to CMC: - Determination of biochar moisture, on-site records - Verification of mass of dry biochar shipped to end users and end use applications - Confirmation of LCA boundary – feedstock processing (E_{biomass})	

The following operations at the Greenville facility were observed and assessed during verification and site audit:

- Receipt of biomass from regional lumber operations, with documentation of feedstock delivered quantity, and acceptance of waste wood;
- Woody biomass from lumber operations as waste wood conveyed to the pyrolysis process;
- Operation of the pyrolysis reactor, including:
 - Pyrolizer input (wet biomass input and quantification);
 - Production and combustion of syngas;
 - Collection, cooling of biochar;
- Storage and packaging of biochar;
- Sale and shipping of biochar to end users;
- Documenting all process operations, including input and output rates, biochar production and product sales, and amounts of by-products.
- Sampling and analysis of the biochar product.

Photographs of the facility and operations captured during the production facility site visit are provided in Appendix 2.

Verifiers also reviewed the processes and calculations used for the LCA and CORC quantifications. CMC utilizes the CarbonPilot software platform (<https://carbonpilot.com/>) to record, track, and report the Greenville facility biochar production, CORCs, and LCA emissions. Verifiers were granted online access to the CMC CarbonPilot account to enable review and verification of the facility inputs, data processing, and platform outputs.

3.2. PROCESS INPUTS & OUTPUTS

The CMC process uses waste biomass as its primary feedstock, with air input to support combustion of gasification products for internal heating, emission control via oxidizer combustion of residual syngas, and production of biochar. Energy-related inputs include fuel (diesel) for materials handling equipment, electricity for systems operation, and fuel (waste biomass and propane) for pyrolysis startup. The

process uses water for biochar handling safety and moisture content management of final product. This water is contained in the final product (there is no process water waste stream). The production and LCA data verified here includes sporadic process operations during the reporting period of 08-14-2023 to 08-18-2024 for a total of 3818 hours of production. Table 3 summarizes the observed inputs and outputs from the process and typical rates from supplied operational data.

Table 3. Verified Production Facility Inputs & Outputs

Input/Output	Verified Rate	Notes (Specifications, source, etc.)
Waste Biomass input (forest and lumber mill woody waste)	Average 6.7 tonne wet biomass input/dry tonne biochar	Hardwood lumber mill waste, chipped to 2-inch size, from regional lumbering operations.
Biomass supply inputs (collection, handling, transportation emissions), (E _{biomass})	130 kg CO ₂ -eq. /dry tonne biochar	Emissions are from the transport of biomass feed to facility. Average roundtrip transport distance is 110 mi.
Production inputs: - diesel (material handling equip) - site electricity use - propane (startup)	Average 5.9 gal/operational day Average 100 kWh/operational day Average 109 gal/startup event	On-site diesel filling station meter used to record weekly loader consumption. Daily power consumption value estimated from equipment specifications and historical utility meter readings. Measured average propane used for process startup and stabilization.
Production water input	Not recorded at the time of the audit	Water use for biochar quench as needed, is estimated to be equivalent to the moisture content of the biochar. Emissions associated with estimated annual water use (30 cubic meters) are negligible and not currently included in E _{production} .
Biochar production output	130.4 dry tonne produced, 45.0 dry tonne sold for end use	Biochar production and distribution during reporting period. Sold biochar end use is direct soil amendment or co-amendment
Production and operation emissions output (E _{production})	180 kg CO ₂ e/tonne biochar DMB	Emissions from all production process and supporting equipment, including process startup, burnout, process operations, syngas destruction, material handling equipment fuel usage, packaging, and equipment manufacture.
Product distribution emissions output (E _{use})	250 kg CO ₂ e/tonne biochar DMB	Verified based on actual transport distances from production facility to purchasers. Assumes end use emissions are included in material application baselines (biochar is comingled with baseline applicants).

3.3. VERIFIED OUTPUT & CORCS

Table 4 includes the specific CORCs claimed by CMC at its Greenville facility for the specified reporting period, and the values verified by 350Solutions during the on-site audit and data review.

TABLE 4. VERIFIED CORCS FOR THE CMC GREENVILLE FACILITY

Performance Metric Name / Description	Claimed Value ¹	Verified Value	Data Source	Reporting Period
CORC Factor	2.54 tonne CO ₂ / tonne biochar	2.54 tonne CO ₂ / tonne biochar	- CMC LCA Projections Report_Final.docx - CMC GHG LCA Workbook updated_Final.xlsx - CORC Report Summary - Biochar_Final_v2.xlsx - CarbonPilot CMC account	08/14/23 to 08/18/24
Biochar Production	45.0 dry tonne	45.0 dry tonne		
Total CORCs	114.3 CORCs	114.3 CORCs		

¹ Claimed values are those submitted by CMC after completion of LCA revisions based on results of Puro reviews and independent audit. Verified values are based on verification of final biochar production records for the reporting period. Output is reported in drv biochar tonnes in accordance with Puro guidance.

4. AUDIT FINDINGS

4.1. SUMMARY OF AUDIT FINDINGS

350Solutions has reviewed and audited the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the Puro Audit Package and during the site visit and follow up:

- ☐ **Meets the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3**
- ☒ **Meets the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3 with minor modifications**
- ☐ **Does Not Meet the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3**

A summary of specific findings associated with each requirement of the Puro Standard and Biochar Methodology and any identified issues with the audit are summarized in Table 5 below.

TABLE 5. AUDIT FINDINGS

Puro Standard Biochar Method. Section Ref.	Audit Verification Topic	Findings
1.1.1 5.2.1	<i>Sustainable Feedstock</i>	Acceptable. CMC obtains waste forest biomass from local logging and lumbering process facilities. Under the baseline alternative, the woody waste would typically be left to rot at the facilities, utilized as furnace fuel, or landfilled.
1.1.2 5.4.2	<i>Biochar Use</i>	Acceptable. CMC has demonstrated that the biochar is sold to a variety of customers using the product as a soil additive or enhancement. Transport emissions associated with biochar transport and delivery are considered in the LCA.
1.1.3 5.2.2 5.3.3 5.4.1	<i>Net-Negative LCA</i>	Acceptable with modifications. CMC has demonstrated an appropriate basis for CORCs according to the Puro Biochar Methodology. CMC utilizes the CarbonPilot software platform to record, track, and process data supporting calculations used for the LCA and CORC quantifications. Opportunities for improvements and modifications for collection and recording of supporting data are provided in Section 4.2 below. The supplier does not currently provide numerical evidence demonstrating that plant maintenance and end-of-life emissions are below the chosen LCA cutoff. The LCA does not currently include emissions associated with application of biochar by end users where applicable (in this case, land applications of biochar are generally co-mingled with existing land spreading materials and therefore associated emissions are zero).
1.1.4	<i>Prohibition of Fossil fuel use for process heat</i>	Acceptable. Process heat is provided by internal combustion of biomass and biomass-derived syngas. Propane is only used for process startup and transition to steady operation. Propane consumption is not metered at the kiln level but estimated based on bulk storage tank readings.
1.1.5	<i>Negligible methane emissions</i>	Acceptable. Emissions from the process only occur from the oxidizer exhaust. The oxidizer operates at conditions such that low methane emissions are evident. Total hydrocarbon (THC) emissions were measured during a regulatory stack test for the State of Maine and currently quantified at 0.03 lbs/hr and included in the LCA (speciated methane emissions are a subset of the THC value).
1.1.6, 5.3.4	<i>Molar H:C Ratio < 0.7 & Biochar Qualities</i>	Acceptable. Biochar analytical results have demonstrated an average molar H:C ratio of 0.17 (std dev 0.07) over six samples analyzed during the reporting period.
1.1.7	<i>Safe Environment & Biochar Handling</i>	Acceptable. CMC operates under an air permit issued by the State of Maine Department of Environmental Protection – Permit A-1164-71-A-N, expiration 10-25-2032. Produced biochar is quenched, collected in super sacks, labeled, and stored on-site until sale and delivery.
1.2.2	<i>Environmental & Social Safeguards</i>	Acceptable. CMC operates under air permits discussed above and written internal safety and operational procedures. Community and stakeholder outreach activities are documented.
1.2.3	<i>Demonstrated Additionality</i>	Acceptable. CMC has submitted that: The project is not required by existing laws, regulations, or other binding obligations. Additional revenue derived from CORC sales will support CMC by providing needed revenue to improve the system and company financials. Without biochar production, the feedstock used at CMC currently would otherwise decompose at the supplying lumber facility.
1.2.4 5.3.1 5.3.2	<i>Biochar Quantification</i>	Acceptable with modifications. The final packaged biochar dispatched from the facility to end users is measured at the facility using a hanging supersack weigh scale. Scale calibration is checked using a non-traceable weight, and cross checked with a second scale on handling equipment. A traceable standard calibration weight or traceable calibration certificate is required for future product mass

		quantification. All product amounts included in inventory and dispatch sales are recorded. CMC also documents biochar and other product production in daily production logs. The LCA inputs are complete.
1.2.5	<i>Verified Production Facility standing data</i>	Acceptable. Conformance to the requirements of Section 1.2.5 of the Biochar methodology is documented and verified.
5.2.1	<i>Feedstock Sustainability Proof</i>	Acceptable. CMC has demonstrated that its feedstock is solely waste wood from regional lumber processing and sawmill facilities, which are otherwise destined for natural decomposition or incineration. Markets for alternative uses of vast amounts of waste wood are shrinking regionally. Shipping documents from the biomass suppliers and transporters and weigh scale records are provided as evidence.
5.4.3	<i>Justification of Soil Temperature</i>	Acceptable. An appropriate regional soil temperature of average 9°C in the project area was used in the LCA. Future applications of end use may occur in warmer regions.
5.5.2	<i>Statement re: Double Counting</i>	Acceptable. CMC is aware of end-user customer practices and assures no-double counting is taking place.
5.5.3	<i>Marketing / Branding Restrictions on end-user</i>	Acceptable. Verifiers have found no evidence of double counting. Producer documents that purchaser assures that they or the end-user will not use biochar for energy production or combustion of any kind and will make no claim to the carbon sequestration from the use of the product.

Verifiers also reviewed the processes and calculations used for the LCA and CORC quantifications. CMC utilizes the CarbonPilot software platform (<https://carbonpilot.com/>) to record, track, and report the Greenville facility biochar production, CORCs, and LCA emissions. The platform is designed to automate Measurement, Reporting, and Verification (MRV), certifications, and product sales for participating production facilities. Verifiers were granted online access to the CMC CarbonPilot account to enable review and verification of the facility inputs, data processing, and platform outputs. Additional details regarding audit activities, documents reviewed, and observations during the audit process are summarized in Section 4.2 below and in Appendix 1.

4.2. AUDIT ISSUES & RECOMMENDATIONS

Findings for the verification of the production facility and the output audit include: (a) audit issues that identify findings that must be addressed to fully conform to the requirements of the Puro standard and biochar methodology, and (b) identification of opportunities for improvement that are recommended to improve future MRV processes. Key findings are summarized in Table 6. Those and other MRV recommendations are detailed in Sections 4.2.1 and 4.2.2 below.

TABLE 6. AUDIT ISSUES AND RECOMMENDATIONS

Finding Category	Finding Description	Recommended Corrective Action	CMC Implementation Status
Audit Issue	The Puro Biochar Methodology mandates calibrated load cell measurements for biochar production quantification.	Procure services for traceable calibration of biochar weigh scale or procure range appropriate calibration standards for in-house calibrations.	CMC has initiated the process of procuring a new digital scale for mass biochar measurements that will include a traceable calibration certification.
	The Puro Biochar Methodology mandates calibrated instrumentation for determination of biochar moisture content.	Procure services for traceable calibration of moisture analyzer balance or procure range appropriate calibration standards for in-house calibration.	CMC has included procurement of a new moisture analyzer with calibrated balance in their 2025 operating budget and plans to implement use of the instrument during that reporting period.
Opportunity for Improvement	Electricity consumption of the biochar production process is currently estimated and not directly measured or recorded.	Install appropriate metering equipment to facilitate measurement and reporting of electricity consumption.	CMC is consulting with an electrical contractor about implementation of a power meter isolated to biochar production processes only. No action has been implemented to date.

4.2.1. Audit Issues

Verifiers reviewed the findings of Puro’s internal review of the original CMC audit package submission and LCA calculations. Significant CMC submittal issues and required action items were identified by Puro. As such, these issues were reviewed during this audit to verify that items were addressed and rectified as necessary by CMC. Corrections were implemented by CMC in the form of modified iterations of documents, LCA assumptions and calculations, and CORC calculations. These included:

- **Environmental Evaluation Report:** The Environmental Evaluation Report was edited, resubmitted, and verified as complete.
- **MRV for Quantification of dry biochar produced:** Procedures and protocols for quantification of biochar production on dry basis have been revised and submitted. Certain MRV parameters of note include:
 - Biochar production rate – Biochar production is reported as a function of the mass of biochar collected in super sacks (wet basis). Each super sack is weighed using an acceptable weigh scale (RESHY Digital Crane Scale 1000Kg Digital Hanging Scale). Scale accuracy is assessed by operators using a standard weight (not metrologically traceable) and by cross checking against the scale on the material handling tractor (also not calibrated). For future audits, the facility must have the primary super sack hanging scale calibrated against metrologically traceable standard weights. Plans are in place to address this finding.
 - Wet basis super sack weights are corrected to dry mass basis using on-site biochar moisture determinations. Sampling and analytical procedures are conducted according to recommended guidance [3], **but the balance used for weighing the biochar samples does not have a traceable calibration.** Traceable balance calibration should be obtained, or a replacement balance with a traceable certification of calibration should be procured for this determination. Plans are in place to address this finding.

- Biomass input records from the scale-equipped loader tractor and biochar output records from the super sack hanging scale are now input to the CMC's CarbonPilot account using an operator employed mobile app.
- Other opportunities for MRV improvements are provided in Section 4.2.2.
- **LCA errors and omissions:** Several omissions and errors were identified in early iterations of the LCA calculations. The current iteration of the LCA is verified to have addressed and included the following findings:
 - Use of fuels on-site contributing to $E_{\text{production}}$ (diesel and propane) are complete, including emissions associated with traceability of UK DEFRA WTT factors for fuels production and supply;
 - Quantification of and emission factors for super sack materials are included in LCA;
 - Methane emissions from syngas combustion are properly documented;
 - Equipment lifetimes updated as 10 years;
 - Using the CarbonPilot software tool, referenced emission factors, and complete inclusion of all required LCA inputs, verifiers have confirmed that the current LCA iteration used to derive CORCs during the reporting period are complete.
- **CORC Summary Report – Quantification of CORCs:** The CORC summary report was revised and verified to rectify the following omissions and inconsistencies noted during the Puro reviews:
 - Biochar characterization of the carbon content and H:C ratio is reported from samples relevant to the reporting periods and verified;
 - Biochar moisture levels are reported based on samples relevant to the reporting periods and determined on-site according to the sampling protocol (see comments above for more detail).
- **Documentation of biochar end use and required analysis:** Biochar end use is currently documented in production logs entered into the CarbonPilot software, as well as customer invoices that document the customer, location, and mass of biomass shipped. **Because biochar analytical requirements may differ depending on the end use application, the intended end use application of each customer shipment should be documented in future reporting periods to ensure that all required biochar analyses are completed.**

For this reporting period, all CMC biochar sold is claimed to be intended for end use as soil amendments. As such, the analyses conducted and results reported are conformant to the Puro rules (verified as meeting the US regulation Code 336 requirements, which are applicable for soil amendments seeking reimbursement for USDA conservation practices and which serve as the local regulations). Additional analyses may be required should other end uses of CMC biochar be applied in future sales (such as IBI Type B testing for animal feedstocks – where, in accordance with section 1.1.7 of the Puro Biochar Methodology, suppliers must screen the biochar

produced to meet the requirements of IBI, EBC, or another standard approved by Puro outlining biochar quality requirements).

4.2.2. Opportunities for Improvement

Recommendations for improving the quality of data and accuracy of the LCA and CORC claims in the future include:

- **Prioritizing the recommended improvements to quantification of dry biochar produced specified in Section 4.2.1 above** (scale calibration for biochar mass and balance calibration for moisture biochar determinations).
- **Monitored water use:** There is no monitoring of water consumption used for quenching biochar and cooling certain pyrolysis system components. Although this is an omission from the Puro methodology requirements, LCA emissions associated with water use are estimated (based on moisture content of biochar product) and actual use is minimal. **However, this lack of data could lead to an underestimation of the environmental impact related to water use and implementing a measurement of water use is recommended.**
- **On site biochar moisture determinations:** In addition to calibration of the balance used for wet and dry measurement of biochar sample mass, a moisture analyzer such as Ohaus MB32 or equivalent is recommended. These instruments, designed for analyses using ASTM D1762-84 standard test method for chemical analysis of wood charcoal would add reliability and credibility to the on-site moisture determinations.
- **Energy use MRV improvements:** **Consumption of diesel fuel for materials handling, propane for pyrolysis startup and stabilization, and electricity use for operation of the pyrolysis system and supporting infrastructure (all components of $E_{\text{production}}$), are currently not directly measured.** To improve LCA accuracy and prevent over reporting of emissions due to conservative energy use estimates, it is recommended that the use of each energy source be measured and recorded for each weekly operating period. Bulk storage gauges for diesel and propane currently in use at the site are sufficient to create these weekly records. A second power meter specific to the biochar process energy use may be required to isolate and record the power consumption. These weekly energy use records can then be normalized to production rates during a given reporting period (e.g., unit of energy consumed/dry tonne biochar produced).
- **Oxidizer Exhaust Emissions:** A full regulatory emissions test of the system exhaust gas was conducted in October 2023. Emissions of all required pollutants were documented as well below the respective limits and there, it is not anticipated that the Maine DEP will require another test in the foreseeable future. As an indication of continued regulatory compliance, it is recommended that the facility continuously monitor and record oxidizer and exhaust stack temperature at a minimum, to demonstrate that operating conditions are consistent with those recorded during the compliance testing.

5. ELIGIBILITY FOR ONGOING ISSUANCE RIGHTS

Puro.earth are currently transitioning to use the 4.0 version of the Puro General Rules. Although the CMC facility and output audits described in this report were conducted using version 3.1 of the General Rules, certain rules described in the updated version of the document (v4.0), such as the Ongoing Issuance Right, are applicable to all projects currently registered or in the process of getting registered on the registry. Specifically, Appendix A of the updated rules dictates that “the evaluation of the Ongoing Issuance Right is done in the performance verification by the 3rd party Auditor as part of the Output Audit” (A.4.1) and that “This evaluation can be done when a Production Facility has demonstrated regular industrial operation and successfully completed performance verification for the previous Monitoring Period with a minimum of 3 months of Output” (A.4.2).

Based on the operational history of the CMC Greenville facility, stability of operations, and production data submitted by CMC, 350Solutions confirms that the facility is able to maintain regular industrial operation and repeat the procedures for regular output monitoring. Additionally, CMC has confirmed plans to address each of the audit issues and opportunities for improvement identified in Section 4.2 above during the next operational year. Many of these MRV upgrades and improvements, when coupled with CMC’s continued utilization of the CarbonPilot platform to record and track production and emissions, will be critical to CMC’s gradual transition to full dMRV and digital real-time monitoring of performance in the future.

350Solutions has reviewed and audited the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the CMC Greenville facility:

- ☒ ***Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A***
- ☐ ***Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A with minor modifications***
- ☐ ***Does Not Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A.***

REVISION HISTORY

Version	Date Issued	Noted Changes
Draft Version (v1.0)	10-08-2024	NA
Draft Version post-QA (v1.1)	10-23-2024	Minor edits and clarifications
Draft Version post-CMC Review (v1.2)	10-25-2024	Minor edits and clarifications and addition of verifier evaluation of Ongoing Issuance rights
Final Version 2.1	10-30-2024	Incorporates review comments from Puro.Earth



AUDITOR SIGNATURES

Bill Chatterton

Bill Chatterton (Lead Auditor)
Carbon Removal Verification Engineer
350Solutions, Inc.

Tim Hansen (Quality Assurance)
Carbon Removal Verification Engineer
350Solutions, Inc

REFERENCES

- [1] Puro.Earth, "Puro Standard General Rules," Version 3.1, [Online]. Available:
<https://puro.earth/documents/>.
- [2] Puro.Earth, " Biochar Methodology," Edition 2022 V3, Feb 2024. [Online]. Available:
<https://puro.earth/methodologies/>.
- [3] Puro.Earth, "Guidance for biochar suppliers on determining the dry mass of biochar produced and delivered," 2022. [Online].

APPENDIX 1: PURO.EARTH BIOCHAR METHODOLOGY AUDIT CHECKLIST

Production Facility and Output Audit - Biochar Methodology	
Audit ID	350PU-CMC-01
Audit Inception Date	19 Sept 2024
Production Facility ID	CMC Biochar
Production Facility Location	Greenville Steam Rd, Greenville, ME, 04441
Auditing Body	350Solutions, Inc.
Auditor Initials	BC
QA	TH

Guideline Ref	Requirement	Requirement Met Y/N	Verification Remarks Insert auditors comments	Evidence Document Insert evidence used to verify requirement
1.2.5	A certified trade registry extract for the CO2 Removal Supplier	Y	CMC provided their articles of conversion effective 04-13-2020	Articles of Conversion CMC RS, Puro_Registration_643002406801000909.pdf
	CO2 Removal Supplier registering Production Facility	Y	CMC registered the Production Facility at Greenville Steam facility with Puro	Signed Puro registration
	Evidence of the location of the Production Facility	Y	185 Greenville Steam Rd, Greenville, Maine 34°22'45"N, 081°46'07"W	Signed Puro registration, site visit and audit
	Evidence of the Volume of Output for the full calendar year prior to registration	Y	Data provided for August 14, 2023 - August 18, 2024, which is the reporting period specified	CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CORC Report Summary - Biochar_Final_v2.xlsx
	Evidence of the Removal Method(s) for which the plant is eligible to receive CORCs	Y	Biochar production from waste wood is an acceptable removal method for receipt of CORCs	CMC LCA Projections Report_Final.docx, site visit and audit
	If the Production Facility has benefited from public support, evidence to show this	Y	No benefit from public support	N/A
	Documentation on Environmental and Social Safeguards imposed	Y	Air quality permit from the state of Indiana and documentation in support of biochar from NRCS were provided.	A1164AN.pdf (air permit) Report - CMC Air Emissions Test Report -Oct 2023.pdf Environmental Evaluation Report_CMC_updated July 2024.pdf GreenvillePermitLetter.pdf
5	Proof of sustainability of raw material for forest biomass (FSC, SFI, PEFC, other certifications)	Y	Vast sources of waste woody biomass in production facility region	Biomass types and origins list _ CMC.xlsx Letter from Feedstock Supplier.pdf Summary of Forest Management in Maine.pdf Puro additionality questions to suppliers v1.8
	Proof of sustainability of raw material for waste biomass	N/A		
	LCA data for biomass and biochar production, supply and use, including climate change impact and the contribution of each life cycle stages.	Y	LCA is complete with Estored, Ebiodmass, Eprod, and Euse calculated separately. No CH4 or N2O emissions expected. Emissions factors used include non-CO2 greenhouse gases to determine CO2-eq	CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CORC Report Summary - Biochar_Final_v2.xlsx CMC Carbon Pilot platform account
	Proof of product quality: laboratory analysis of total organic carbon content, hydrogen content and H/Corg	Y	Control Labs is a widely used biochar analysis lab	3060579-01 IBI Biochar Report.pdf 3070467-01 Proximate Biochar Report.pdf
	Proof of production volume: documentation for the whole period and methodology applied to calculate the dry mass of biochar produced.	Y	All material was bagged during this reporting period. Each bag was weighed using a loadcell under the rack that holds the supersack	CORC Report Summary - Biochar_Final_v2.xlsx CMC Carbon Pilot platform account Inspection of operator field logs

	For mobile units or carbonizer operator: proof of load cell measurement of the biochar for the whole period, and water input measurement.	N/A		
	Proof of end use of biochar: offtake agreement, shipment, and other records indicating the intended use of biochar.	Y	Biochar is used for application on farm fields or for other soil enhancements. Customers submitted letters stating they would not burn the material.	End user invoices.pdf
	Justification on the soil temperature selected for the calculation of the biochar sequestration.	Y	Biochar is mostly used in Maine and New England region. The average temperature over one year is 9.4 degrees C. Global datasets for annual average soil temperature (at depth of 5-15 cm) used.	Soil temperature calculation and justification, https://onlinelibrary.wiley.com/doi/10.1111/gcb.16060
	Proof of no double-counting	Y	No C-negative branding used by CMC and no attempts to monetize the carbon credits except through Puro	CMC usage and marketing statement
	Proof of sales	Y	Copies of sold product invoices provided	Invoice **.pdf (39 total)
1.1.1	Biochar is used in applications other than energy.	Y	Biochar applied to fields or used as soil enhancement or amendment	CMC usage statement and end user invoices
1.1.2	Biochar is produced from sustainable forest or waste biomass raw materials (consult list of raw materials).	Y	Biochar is produced from waste forest wood that would otherwise decompose	Puro additionality questions to suppliers v1.8, feedstock delivery and invoicing records
1.1.3	LCA shows:- carbon footprint of the biomass production and supply- emissions from the biochar production process- carbon footprint of the biochar end use- cradle to grave	Y	Biomass is a waste material and is appropriately placed outside the system boundary; Biomass transportation is accounted for using an average distance traveled; Biochar production emissions from propane, diesel, and electricity consumption is conservative; Emissions from packaging material (supersacks) production are included; Infrastructure and equipment manufacturing are accounted for.	CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CORC Report Summary - Biochar_Final_v2.xlsx CMC Carbon Pilot platform account
1.1.4	Pyrolysis reactor input fuel for heating is not a fossil fuel. Unless only used for ignition/pre heating or in a mobile unit and the emissions are fully included in the LCA. The use of waste heat from other industrial processes (eg. Biodigesters, cement production) is permitted.	Y	Propane is only used for cold start-up ignition and kiln draft stabilization and is accounted for in the LCA	CMC GHG LCA Workbook updated_Final.xlsx CORC Report Summary - Biochar_Final_v2.xlsx CMC Carbon Pilot platform account
1.1.5	Pyrolysis gases are combusted or recovered. Bio-oil and pyrolysis gases can be stored for later use as renewable energy or materials.	Y	Produced gases are combusted in thermal oxidizer attached to gasifier.	CMC GHG LCA Workbook updated_Final.xlsx CORC Report Summary - Biochar_Final_v2.xlsx CMC Carbon Pilot platform account Site visit observation
1.1.6	The molar H/Corg ratio is less than 0.7.	Y	Average H/Corg during production period = 0.17	3060579-01 IBI Biochar Report.pdf 3070467-01 Proximate Biochar Report.pdf

				CORC Report Summary - Biochar_Final_v2.xlsx
1.1.8	Evidence of safe handling and transport is provided and adequate for the production facility.	Y	Biochar itself is handled safely. It is moved using heavy equipment, not manually	Site visit
1.1.7	Biochar products meet the product quality requirements existing in the jurisdiction where biochar is used and for the specific applications considered.	Y	No relevant product quality requirements exist in the jurisdiction, so the IBI standard defines the requirements. Biochar is acceptable for use according to USDA's BioPreferred Program and the NRCS. The biochar has not been tested for the organic contaminants named in IBI's "Type B" tests.	3060579-01 IBI Biochar Report.pdf
6.6 in General Rules	The SDG outcome meets the claims below: Specific: must refer to a specific individual outcome rather than being overly broad or abstract. Cause-and-effect relationship: must establish a direct and inherently clear cause and-effect relationship between the carbon removal activity and the impact, i.e., the impact should be additional and attributable to the CO2 Removal activity. Quantitative metric: Numeric measurement allows for objective assessment and comparison as per Section 4 of SDG assessment requirements Accuracy: can be determined without relying heavily on input assumptions. Manageable MRV: Pragmatic MRV processes limit the costs of data collection.	Y	Biochar production at CMC Greenville is only possible with carbon finance and was not in frequent operation until manure became an acceptable biomass feedstock. Few input assumptions are needed to quantify CDR, rather empirical measurements are used.	Puro additionality questions to suppliers v1.8 CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CMC Carbon Pilot platform account Clean Main Carbon - Cost Analysis - Financial Additionality (5.3.2023) (3).xlsx
1.2.1	Evidence of Production Facility eligibility under the general rules of Puro Standard.	Y	See Standing Data Confirmation section, above	Puro additionality questions to suppliers v1.8
1.2.2	The Production Facility demonstrate Environmental and Social Safeguards.	Y	See Standing Data Confirmation section, above	A1164AN.pdf (air permit) Report - CMC Air Emissions Test Report -Oct 2023.pdf Environmental Evaluation Report_CMC_updated July 2024.pdf GreenvillePermitLetter.pdf
1.2.3	CO2 Removal Supplier shall be able to demonstrate additionality, meaning that the project must convincingly demonstrate that the CO2 removals are a result of carbon finance. Even with substantial non-carbon finance support, projects can be additional if investment is required, risk is present, and/or human capital must be developed. To demonstrate additionality, CO2 removal Supplier must provide full project financials and counterfactual analysis based on	Y	CMC has demonstrated conformance with these requirements.	Puro additionality questions to suppliers v1.8 CMC LCA Projections Report_Final.docx Clean Main Carbon - Cost Analysis - Financial Additionality (5.3.2023) (3).xlsx Additionality Form Signed.pdf

	Baselines that shall be project-specific, conservative and periodically updated. Suppliers must also show that the project is not required by existing laws, regulations, or other binding obligations.			
1.2.4	The Production Facility's documentation system is accurate and reliable	Y	Followed example biomass and biochar delivery from input to database during site visit	Site visit, operator interviews CMC Carbon Pilot platform account Operational field logs and records
	The quantity of the biochar produced and sold is quantified and documented in a reliable manner	Y	Invoices provided; demonstrated invoice production during site visit. On-site biochar weigh scale to be calibrated to traceable standard.	Site visit, operator interviews Operational field logs and records Customer invoice **.pdf (39 total)
	Relevant meters are in place and they are calibrated;	Y	Daily average electricity, diesel, and propane usage based on site meter readings. On-site biochar weigh scale to be calibrated to traceable standard.	Site visit, operator interviews Operational field logs and records
	The emissions from the cultivating, harvesting and transporting of the biomass are estimated and calculated in a reliable manner	Y	Biomass is a waste material so emissions associated with production are outside of the LCA scope. Transportation emissions are accounted for by using the tonnes of biomass delivered and the average delivery distance to facility	CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CMC Carbon Pilot platform account Site visit, operator interviews Operational field logs and records
	The energy use of the Production Facility can be quantified and the emissions from the process calculated	Y	Daily average electricity, diesel, and propane usage based on site meter readings. All energy use attributed to biochar production. EPA emissions factors used.	
	The auditor goes through the Quantification of CO2 Removal requirements with the CO2 Removal Supplier, so that the Supplier is able to calculate the CO2 Removal independently in its Output Report	Y	CMC can independently calculate CORCs, utilizes CarbonPilot software platform, https://carbonpilot.vercel.app/company/output-report/overview	
3.1	The CO2 Removal Supplier shall provide a life cycle assessment (LCA) for biochar activity including disaggregated information on the emissions arising at different stages. The system boundary is set cradle-to-grave and shall include emissions from production and supply of the biomass, from biomass conversion to biochar, and from biochar distribution and use.	Y	Logging woody feedstock is a waste biomass and thus the production can be excluded from the LCA system boundary (Fig 2B in Methodology)	CMC LCA Projections Report_Final.docx
3.2	Life cycle assessment (LCA) shall follow ISO standard, WRI GHG protocol or similar method.	Y	The analysis has been conducted and presented consistent with Puro.Earth's Biochar Methodology for quantifying net CO ₂ Removals and guidance from WRI's GHG Protocol.	CMC LCA Projections Report_Final.docx CMC GHG LCA Workbook updated_Final.xlsx CMC Carbon Pilot platform account

3.3	The default baseline emission scenario for the project activity feedstock is zero, which is a conservative assumption since it is not taking into account methane emissions derived from decay of manure or combustion of waste biomass. If a non-zero baseline presented, needs to be accepted by Puro.earth	Y	Baseline emission scenario is assumed to be zero	Puro additionality questions to suppliers v1.8
1.2.2	Environmental Safeguards are demonstrated through: Environmental Impact Assessment (EIA) Environmental permit Other documentation approved by the Issuing Body on the analysis and management of the environmental and social impacts	Y	Environmental safeguards are demonstrated through Environmental Evaluation Report and Maine DEP air permit	A1164AN.pdf (air permit) Environmental Evaluation Report_CMC_updated July 2024.pdf Report - CMC Air Emissions Test Report -Oct 2023.pdf CMC Biochar Safety.pdf
1.1.8	Safety measures include, but are not limited to, providing a Material Safety Data Sheet, post-production quenching and cooling of biochar, and appropriate flue gas treatment systems.	Y	Biochar is quenched to ~20% moisture after production and flue gas is treated using a thermal oxidizer and is ultimately released through a stack. Worker safeguards are in place as standard operating practices. Biochar MSDS and Environmental, Health, and Safety Manual provided.	
6.7.3.1 in General Rules	The risk estimation shall be quantitative, and the Overall Risk Effect shall be given in percentage of the Output volume. The Reversal risk estimation shall consider all material risks arising from: Nature-induced risks (e.g., flora, fauna, or climate conditions). Human-induced risks (e.g., design and construction faults, operational risks). Geopolitical risks (e.g., potential effects of the legal and political environment). Any additional factors mentioned in the applicable Methodology.	Y	No risk of reversal because biochar is applied on fields or as other soil enhancement uses. Customer attestations that the biochar is not burned were provided.	CMC usage statement and end user invoices

Guideline Ref	Requirement	Requirement Met Y/N	Verification Remarks Insert auditors comments	Value	Units
4.2	Qbiochar = Quantity of biochar produced and sold to end user. (dry char)	Y		45.0	tonnes
	FpTHTs = $c + m \times H/Corg$	Y		1.00	N/A
	C Biochar = carbon content of biochar	Y		89.4%	N/A
	Estored = biochar carbon storage = Qbiochar x Cbiocharorg x FpTHTs x 44/12	Y		3,100	kg CO _{2e} /tonne biochar
4.3	Ebiomass = LCA emissions of production and supply of biomass: With allocation	Y		130	
4.4	Eproduction = LCA emissions from biochar manufacturing: With allocation	Y		180	
4.5	Euse = LCA emissions of the use of biochar, including distribution up to the point of final use: With allocation	Y		250	
4.1	CORCs = Estored - Ebiomass - Eproduction - Euse: with allocation	Y		114.3	tonnes
	Quantity of CORCs (in evidence).	Y		114.3	tonnes

APPENDIX 2: SITE VISIT PHOTOS



Figure A2-1. CMC 2-inch Hardwood Feedstock Supply



Figure A2-2. CMC Biochar Production Process



Figure A2-3. CMC Biochar Production Process – Kiln Exit



Figure A2-4. CMC Biochar Production Process – Kiln Temperature Control Screen

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications

Company Name:	Clean Maine Biochar	
Date:	9/10/2024	
Verifier Name:	Bill Chatterton and Tim Hansen	
Company Name (where applicable):	350Solutions	
Verifier Contact Information:	bill@350solutions.com , 984-215-0585	
Verifier Address:	1053 E. Whitaker Mill Rd. Suite 115, Raleigh, NC 27604	
Verifier Scope of Activities:	Verification through observation and review of key technology components and documentation.	
Verifier Qualifications	Criteria Met?	Evidence / Notes (note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☒	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures relevant to the Puro.Earth General Standard.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☒	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☒	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☒	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☒	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☒	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Puro Earth CORCs;	☒	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities. These procedures and quality management programs are generally relevant to verification under the Puro.Earth General Standard. Note that verifications completed for Puro.Earth are not equivalent to ISO 14034 verifications.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Puro.Earth for verification, or represent the parties engaged in those activities.	☒	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☒	

William Chatterton
350Solutions, Verification Program Manager

EDUCATION

B.S. Environmental Science, SUNY at Plattsburgh, 1982

Certified Measurement and Verification Professional (CMVP), 2019

Professional Experience

William Chatterton is an Environmental Scientist with 28 years' experience in technology evaluation and demonstration, project management, air pollution monitoring, testing, and regulation. He serves as Program Manager at 350Solutions and manages projects and programs for commercial and government clients. Previously the past 20 years at Southern Research, Mr. Chatterton has managed, and supported programs designed to integrate, demonstrate, and evaluate technology performance in the advanced energy field. Technology demonstrations and evaluations that he has been involved with include technologies designed to promote sustainable energy sources, increase energy use and efficiency, mitigate GHG and other emissions, and in most cases provide other social and economic benefits to potential users. Mr. Chatterton has been heavily involved in the evaluation of numerous emerging energy technologies, distributed generation technologies, and technologies relevant to transportation and oil and gas markets. Mr. Chatterton's roles in support of these projects has included program and project management from administrative and technical perspectives, lead or technical support on test plan development, method development and validation, design and implementation of field-testing activities, data evaluation and presentation, and reporting of results. He has managed numerous projects for both commercial and government clients.

350Solutions: 08-2019 – Present

Verification Program Manager: As Verification Program Manager, Mr. Chatterton manages and executes technology performance demonstrations and verifications of emerging energy (efficiency and green building) and transportation technologies, primarily for U.S. governmental agencies, energy research associations, and state energy agencies. These performance evaluations generally involve evaluation of commercial feasibility, economic impacts (installation, operating, and capital costs, simple payback, and return on investment), environmental impacts (primarily greenhouse gas and criteria pollutant emission reductions), and technology performance. He also manages and monitors 350Solutions' quality management programs and ISO accreditations.

Southern Research Institute: 1999 - 2019

Program Manager, Energy & Environment Technologies: As Program Manager, Mr. Chatterton has managed and executed several technology performance demonstrations and verifications of emerging energy (efficiency and green building) and transportation technologies, primarily for U.S. governmental agencies, energy research associations, and state energy agencies. Mr. Chatterton also has direct experience with management and execution of projects under DOE and DoD grants and contracts. He has recently managed activities on three large DoD projects including Demonstration of a Solar Thermal Combined Heating, Cooling and Hot Water System Utilizing an Adsorption Chiller for DoD Installations, Demonstration and Verification of the Performance of Microturbine Power Generation Systems Utilizing Renewable Fuels, and the Electric Power with Small Scale Organic Rankine Cycle (ORC) Engine/Generator Technology demonstration.

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:

B.S., Chemical Engineering, University of Virginia, 1993

M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:

Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:

2019-Present Founder – CEO, 350Solutions, Inc.

Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research

Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from \$25,000 to \$6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research

2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research

1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:

Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a \$20M prize competition for technologies that capture and beneficially utilize CO₂. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

350Solutions, Inc. Corporate Experience

350Solutions serves as an independent expert in cleantech, low carbon, and environmental technologies. We provide an unbiased assessment of innovative technologies. 350Solutions is accredited through ANAB under ISO 17020 as an independent inspection body to provide independent technology evaluation services using the ISO 14034 ETV process. In addition, 350Solutions staff include a Certified Measurement and Verification Professional (CMVP for IPMVP) and a North Carolina Registered Professional Engineer (P.E.). 350Solutions ANAB Accreditation certificate is provided below.



CERTIFICATE OF ACCREDITATION

The ANSI National Accreditation Board

Hereby attests that

350Solutions, Inc.
1053 E. Whitaker Mill Rd., Suite 115
Raleigh, NC 27604

Fulfills the requirements of

ISO/IEC 17020:2012

and

**ISO 14034:2016, Environmental Management - Environmental
Technology Verification (ETV)**

In the field of

INSPECTION

This certificate is valid only when accompanied by a current scope of accreditation document.

The current scope of accreditation can be verified at www.anab.org.



R. Douglas Leonard Jr., VP, PILR SBU

Expiry Date: 25 September 2024
Certificate Number: AI-2618



An inspection body's fulfilment of the requirements of ISO/IEC 17020:2012 means the inspection body meets both the technical competence requirements and management system requirements that are necessary for it to consistently deliver technically valid inspection results (refer to joint ISO-ILAC-IAF Communiqué dated Sept 2013).



SCOPE OF ACCREDITATION TO ISO/IEC 17020:2012

and

ISO 14034:2016, Environmental Management - Environmental Technology Verification (ETV)

350Solutions, Inc.
1053 E. Whitaker Mill Rd., Suite 115
Raleigh, NC 27604
Tim Hansen tim@350Solutions.com
(919) 675-6432

INSPECTION TYPE A (THIRD-PARTY) BODY

Valid to: **September 25, 2024**

Certificate Number: **AI-2618**

General

Products Categories	Range	Stage	Methods and Procedures
Energy Technologies (ET):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP-350-223-02 - SOP ISO 14034 ETV
Cleaner Production and Processes (CPP):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV
Air pollution monitoring and abatement (APP):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV
Water monitoring and treatment (WMT):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV

Note:

1. This scope is formatted as part of a single document including Certificate of Accreditation No. AI-2618.



R. Douglas Leonard Jr., VP, PILR SBU

Version 005 Issued: August 09, 2022

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